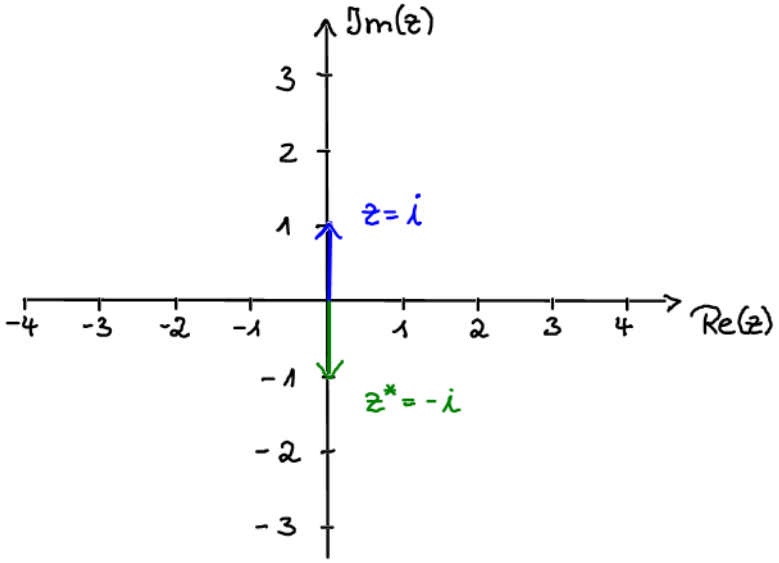
Lösung: Zeichnen und konjugieren

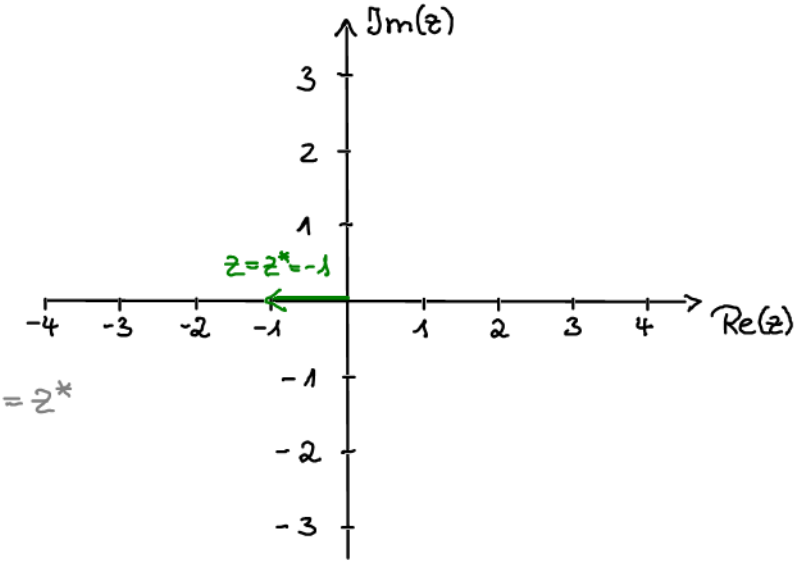
a) $z = i$

$\Leftrightarrow \quad z^* = -i$



b) $z = -1$

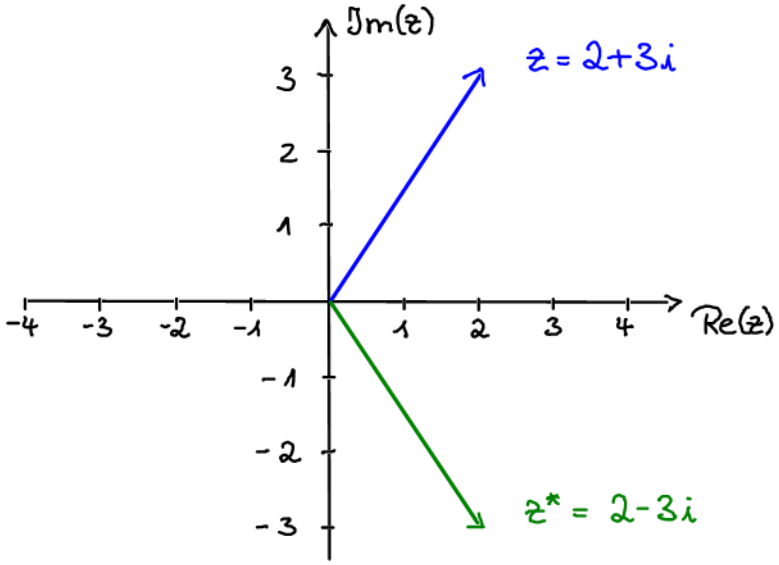
$\Leftrightarrow \quad z^* = -1 \quad (= z)$



es gilt: $z \in \mathbb{R} \Leftrightarrow z = z^*$

c) $z = 2 + 3i$

$\Leftrightarrow \quad z^* = 2 - 3i$



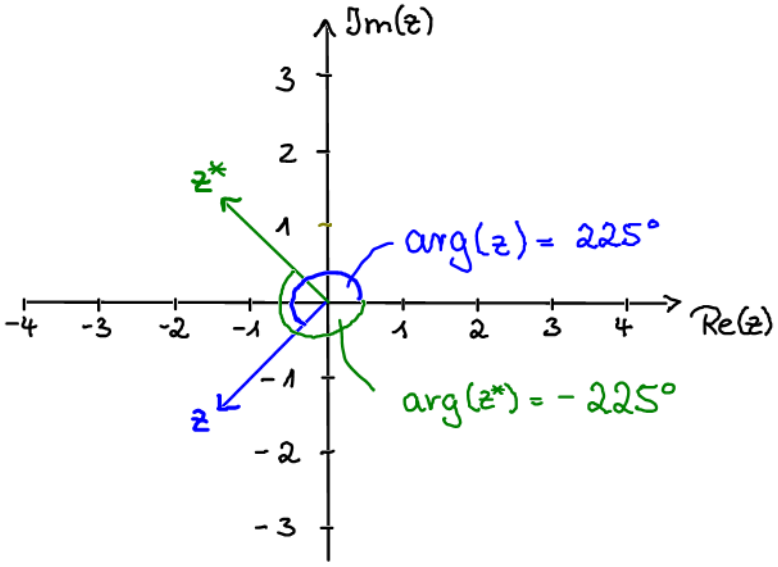
d) $z = 2 \cdot e^{i\frac{5\pi}{4}} \quad \Leftrightarrow \quad z^* = 2\,e^{-i\frac{5\pi}{4}}$

Betrag: $|z| = 2$

Winkel: $\arg(z) = \frac{5\pi}{4} \quad (\hat{=}\ 225°)$

Realteil: $\mathrm{Re}(z) = 2 \cdot \cos\left(\frac{5\pi}{4}\right) = 2 \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\sqrt{2}$

Imag.-teil: $\mathrm{Im}(z) = 2 \cdot \sin\left(\frac{5\pi}{4}\right) = 2 \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\sqrt{2}$



e) $z = 2 \cdot e^{-i\frac{3}{4}\pi} \quad \Leftrightarrow \quad z^* = 2 \cdot e^{+i\frac{3}{4}\pi}$

Betrag: $|z| = 2$

Winkel: $\arg(z) = -\frac{3\pi}{4}$ $(\hat{=} -135°)$

Realteil: $\mathrm{Re}(z) = 2 \cdot \cos\left(-\frac{3\pi}{4}\right) = 2 \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\sqrt{2}$

Imag.-teil: $\mathrm{Im}(z) = 2 \cdot \sin\left(-\frac{3\pi}{4}\right) = 2 \cdot \left(-\frac{\sqrt{2}}{2}\right) = -\sqrt{2}$

